Report

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PKCERT AI AND SOFTWARE DEVELOPMENT INTERNEE

TASK 08

Classification Models:

Logistic Regression, Decision Trees & Random Forests

1. Introduction

The objective of this project is to implement and compare different classification algorithms using the **Diabetes Dataset**. Three machine learning models—**Logistic Regression**, **Decision Tree**, and **Random Forest**—were trained and evaluated to predict whether a patient is diabetic based on medical attributes.

2. Dataset Description

The project uses the **diabetes.csv** dataset.

Features

Feature	**Description**
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Skin fold thickness
Insulin	Serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Genetic diabetes likelihood
Age	Age of patient
Outcome	Target variable (0 = Non-Diabetic, 1 = Diabetic)

3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded dataset using Pandas.
- Checked dataset information.

- Checked missing values.
- Split data into features (X) and target (y).
- Applied train-test split.
- Standardized numerical features (for Logistic Regression).

4. Classification Models

Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Working Principle of Logistic Regression

- Logistic Regression is a **supervised machine learning algorithm** used for **classification problems**, especially when the output has two possible classes (binary classification).
- In this project, Logistic Regression is used to classify patients into two categories:
  0 → Non-Diabetic
  1 → Diabetic
- The algorithm works by learning the relationship between input features and the target variable. It analyzes patient information such as glucose level, BMI, age, blood pressure, and insulin level to predict the probability of diabetes.
- Logistic Regression uses a mathematical function called the **Sigmoid Function** to convert the output into a probability value between 0 and 1.

- The sigmoid function is: $\sigma(z)=1 / 1+ e^{-z}$

Where:

z represents the weighted combination of input features.

e is the mathematical constant.

Logistic Regression is a simple yet powerful classification algorithm. In this diabetes prediction project, it helps predict whether a patient is diabetic or non-diabetic by learning patterns from medical data. Its simplicity, interpretability, and efficiency make it widely used in real-world classification applications.

Importance of each Metric

Metric	Importance
Accuracy	Shows the overall correctness of the model. It is useful when classes are balanced.
Precision	Measures how reliable positive predictions are. Important when false alarms should be reduced.
Recall	Measures the ability to detect actual positive cases. Very important in medical diagnosis because missing a disease case is risky.
F1-Score	Provides a balance between precision and recall and is useful when both types of errors matter.

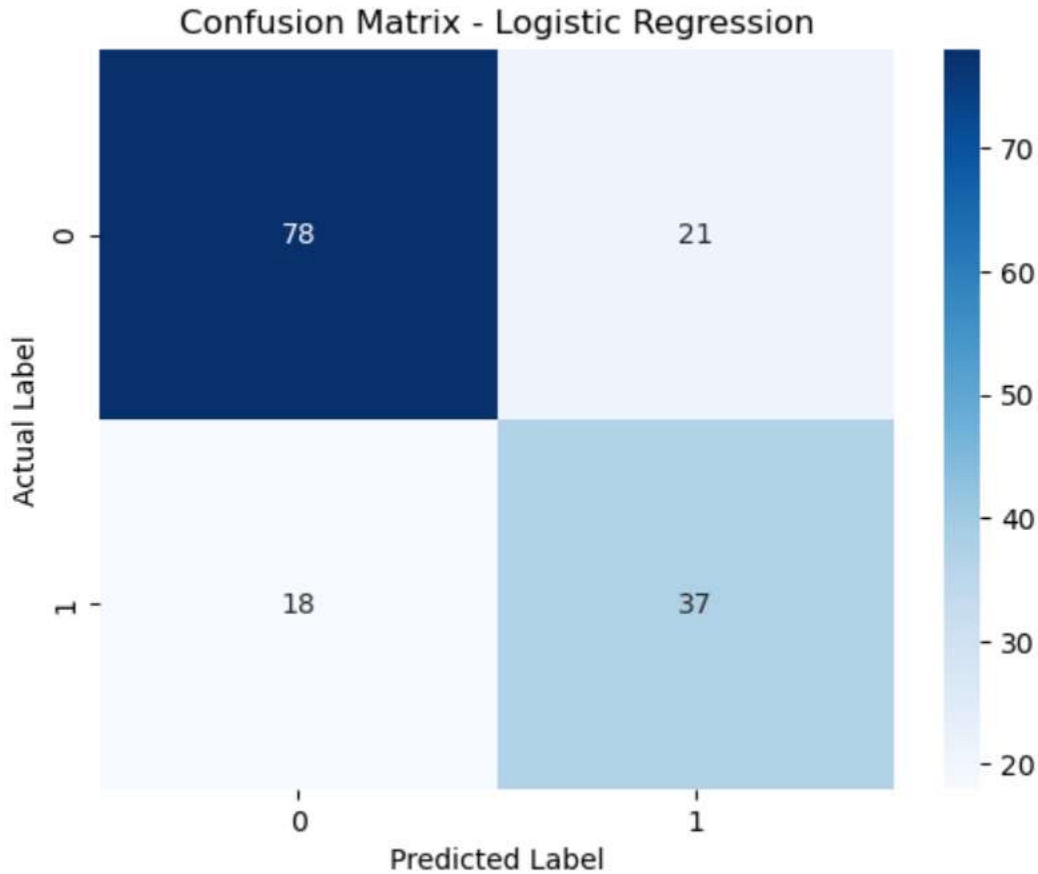
Decision Tree

Decision Tree classifies data by learning decision rules from training data.

Evaluation includes:

- Accuracy
- Classification Report
- Confusion Matrix

A confusion matrix shows the number of correct and incorrect predictions made by the classification mode



Random Forest

Random Forest combines multiple Decision Trees to improve prediction performance and reduce overfitting.

Evaluation includes:

- Accuracy
- Classification Report
- Confusion Matrix

5. Performance Comparison

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.746753	0.637931	0.672727	0.654867
1	Decision Tree	0.746753	0.625000	0.727273	0.672269
2	Random Forest	0.720779	0.607143	0.618182	0.612613

6. Results

Recommended Model: Logistic Regression

Based on the evaluation results, Logistic Regression is the most suitable model for the Diabetes dataset. It achieved the highest precision (0.6379) and highest F1-score (0.6549) among the three models while also achieving the same accuracy (74.68%) as the Decision Tree. Although the Decision Tree obtained a slightly higher recall (0.7273), its precision was lower, indicating a higher number of false positive predictions.

- In medical diagnosis, particularly for diabetes prediction, it is important to have a model that provides a good balance between correctly identifying diabetic patients and minimizing incorrect positive predictions.
- The higher F1-score of Logistic Regression indicates that it offers the best balance between precision and recall.
- Additionally, Logistic Regression is simpler, computationally efficient, less prone to overfitting than Decision Trees, and produces interpretable results, making it a reliable choice for healthcare applications.

The Random Forest model performed worse than both Logistic Regression and Decision Tree in terms of accuracy, precision, recall, and F1-score. Therefore, it is not the preferred model for this dataset.

7. Conclusion

This project demonstrated the implementation of three classification algorithms on the Diabetes Dataset. Among the evaluated models, Logistic Regression produced the best overall performance, making it the most suitable model for diabetes prediction on this dataset.